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5 Techniques of Integration

We discuss a number standard techniques for computing integrals: substitution methods, integration by parts, partial fractions, and improper integrals.

5.1 Basic Antiderivatives

- Here is a list of common indefinite integrals that should already be familiar:

$$\begin{aligned}
\int x^n\,dx &= \frac{x^{n+1}}{n+1} + C,\ n \neq -1 \\
\int x^{-1}\,dx &= \ln(x) + C \\
\int e^x\,dx &= e^x + C \\
\int \sin(x)\,dx &= -\cos(x) + C \\
\int \cos(x)\,dx &= \sin(x) + C \\
\int \sec^2(x)\,dx &= \tan(x) + C \\
\int \sec(x)\tan(x)\,dx &= \sec(x) + C \\
\int \frac{1}{\sqrt{1-x^2}}\,dx &= \sin^{-1}(x) + C \\
\int \frac{1}{1+x^2}\,dx &= \tan^{-1}(x) + C \\
\int \frac{1}{x\sqrt{x^2-1}}\,dx &= \sec^{-1}(x) + C
\end{aligned}$$

- Here are some other slightly more difficult antiderivatives that crop up occasionally:

$$\begin{aligned}
\int \ln(x)\,dx &= x\ln(x) - x + C \\
\int \tan(x)\,dx &= -\ln(\cos(x)) + C \\
\int \sec(x)\,dx &= \ln(\sec(x) + \tan(x)) + C \\
\int \csc(x)\,dx &= -\ln(\csc(x) + \cot(x)) + C \\
\int \cot(x)\,dx &= \ln(\sin(x)) + C
\end{aligned}$$

5.2 Substitution

- The general substitution formula states that $\boxed{\int f'(g(x)) \cdot g'(x)\,dx = f(g(x)) + C}$. It is just the Chain Rule, written in terms of integration via the Fundamental Theorem of Calculus. We generally don't use the formula written this way. To do a substitution, follow this procedure:
  - Step 1: Choose a substitution $u = g(x)$.
  - Step 2: Compute the differential $du = g'(x)\,dx$.
  - Step 3: Rewrite the original integral in terms of u:
    - 3a: Rewrite the integral to peel off what will become the new differential du.
    - 3b: Write the remaining portion of the integrand in terms of u.
    - 3c: Find the new limits of integration in terms of u, if the integral is a definite integral. [If the old limits are $x = a$ and $x = b$, the new ones will be $u = g(a)$ and $u = g(b)$. In a very concrete sense, these are "the same" points.]
    - 3d: Write down the new integral. If the integral is indefinite, substitute back in for the original variable.
- Substitution is best learned by doing examples:
- Example: Evaluate $\int_0^3 2x\,e^{x^2}\,dx$.
  - Step 1: The exponential has a 'complicated' argument x^2, so we try setting $u = x^2$.
  - Step 2: The differential is $du = 2x\,dx$.
  - Step 3a: We can rearrange the integral as $\int_0^3 e^{x^2} \cdot (2x\,dx)$.
  - Step 3b: The "remaining portion" of the integrand is e^{x^2}, which is just e^u.
  - Step 3c: We see that $x = 0$ corresponds to $u = 0^2$ and $x = 3$ corresponds to $u = 3^2$.
  - Step 3d: Putting it all together gives $\int_0^9 e^u\,du = e^u|_{u=0}^9 = \boxed{e^9 - 1}$.
- Example: Evaluate $\int_1^e \dfrac{(\ln(x))^2}{x}\,dx$.
  - Step 1: It might not look like any function has a 'complicated' argument, but if we think carefully we can see that the numerator is what we get if we plug in $\ln(x)$ to the squaring function. So we try setting $u = \ln(x)$.
  - Step 2: The differential is $du = \dfrac{1}{x}\,dx$.
  - Step 3a: We can rearrange the integral as $\int_0^3 [\ln(x)]^2 \cdot \left(\dfrac{1}{x}\,dx\right)$.

- Step 3b: The "remaining portion" of the integrand is $[\ln(x)]^2$, which is just u^2.
- Step 3c: We see that $x = 1$ corresponds to $u = \ln(1) = 0$ and $x = e$ corresponds to $u = \ln(e) = 1$.
- Step 3d: Putting it all together gives $\int_0^1 u^2\,du = \frac{1}{3}u^3|_{u=0}^1 = \boxed{\frac{1}{3}}$.

- Example: Evaluate $\int_0^1 x\sqrt{3x^2+1}\,dx$.
  - Step 1: Here we see that the square root function has the 'complicated' argument $3x^2+1$ so we try $u = 3x^2+1$.
  - Step 2: The differential is $du = 6x\,dx$.
  - Step 3a: We can rearrange the integral as $\int_0^1 \sqrt{3x^2+1}\cdot\frac{1}{6}\cdot(6x\,dx)$. Note that we introduced a factor of $6\cdot\frac{1}{6}$, which is okay since it's just multiplication by 1.
  - Step 3b: The "remaining portion" of the integrand is $\sqrt{3x^2+1}\cdot\frac{1}{6}$, which is just $\frac{1}{6}u^{1/2}$.
  - Step 3c: We see that $x = 0$ corresponds to $u = 1$ and $x = 1$ corresponds to $u = 4$.
  - Step 3d: Putting it all together gives $\int_1^4 \frac{1}{6}u^{1/2}\,du = \frac{1}{6}\cdot\frac{2}{3}u^{3/2}|_{u=1}^4 = \boxed{\frac{7}{9}}$.

- Example: Evaluate $\int_2^3 \frac{2x}{x^2-1}\,dx$.
  - Step 1: Try the denominator: $u = x^2-1$.
  - Step 2: The differential is $du = 2x\,dx$.
  - Step 3a: We can rearrange the integral as $\int_2^3 \frac{1}{x^2-1}\cdot(2x\,dx)$.
  - Step 3b: The "remaining portion" of the integrand is $\frac{1}{x^2-1}$, which is just u^{-1}.
  - Step 3c: We see that $x = 2$ corresponds to $u = 3$ and $x = 3$ corresponds to $u = 8$.
  - Step 3d: Putting it all together gives $\int_3^8 u^{-1}\,du = \ln(u)|_{u=3}^8 = \boxed{\ln 8 - \ln 3}$.
  - Remark: It's possible to do this one without substitution, by using partial fractions to see that $\frac{2x}{x^2-1} = \frac{1}{x+1}+\frac{1}{x-1}$. This gives $I = [\ln(x+1)+\ln(x-1)]\,|_{x=2}^3 = \ln(4)+\ln(2)-\ln(3) = \ln(8)-\ln(3)$ as before.

5.3 Integration by Parts

- The integration by parts formula states $\boxed{\int f'\cdot g\,dx = f\cdot g - \int f\cdot g'\,dx}$. It is just the Product Rule, rearranged and rewritten in terms of integrals using the Fundamental Theorem of Calculus. To perform an integration by parts, all that is required is to pick a function $f(x)$ and a function $g(x)$ such that the product $f'(x)\cdot g(x)$ is equal to the original integrand. Examples will make everything clear.

- Example: Evaluate $\int_0^1 x\,e^x\,dx$.
  - The integrand $x\cdot e^x$ is an obvious product, and so we need to decide which of x and e^x should be f'. Since x gets more complicated if we take its antiderivative (since we'd get $\frac{1}{2}x^2$) we try $f' = e^x$ and $g = x$, to get $f = e^x$ and $g' = 1$. Plugging into the formula gives $\int_0^1 x\,e^x\,dx = x\,e^x|_{x=0}^1 - \int_0^1 1\cdot e^x\,dx = (x\,e^x - e^x)|_{x=0}^1 = \boxed{1}$.

- Example: Evaluate $\int_1^e [\ln(x)]^2\,dx$.

- We can write $[\ln(x)]^2 = \ln(x) \cdot \ln(x)$, but this doesn't help unless we remember the antiderivative of $\ln(x)$. Instead we write the integrand as $1 \cdot [\ln(x)]^2$, so as to take $f' = 1$ and $g = [\ln(x)]^2$. Then we get $f = x$ and $g' = 2 \cdot \ln(x) \cdot \dfrac{1}{x}$ by the Chain Rule. So plugging in will yields

$$\begin{aligned}
\int_1^e [\ln(x)]^2 \, dx &= x \cdot \ln(x)^2|_1^e - \int_1^e x \cdot 2\ln(x) \cdot \frac{1}{x} \, dx \\
&= e - \int_1^e 2\ln(x) \\
&= e - \left[2x \cdot \ln x|_1^e - \int_1^e 2x \cdot \frac{1}{x} \, dx\right] \quad \text{(IBP again)} \\
&= e - [2e - (2e - 2)] \\
&= \boxed{e - 2}.
\end{aligned}$$

- Example: Evaluate $\int_0^1 (x^2 - 2x + 2)e^{3x} \, dx$.

  - We want g to be the thing which gets simpler when we differentiate. The polynomial $x^2 - 2x + 2$ gets much simpler if we differentiate it, while the exponential e^{3x} stays basically the same. So we should take $g = x^2 - 2x + 2$ and $f' = e^{3x}$, so that $g' = 2x - 2$ and $f = \frac{1}{3}e^{3x}$. Then integrating by parts yields an expression which we can't evaluate directly – we have to integrate by parts again:

$$\begin{aligned}
\int_0^1 (x^2 - 2x + 2)e^{3x} \, dx &= (x^2 - 2x + 2) \cdot \frac{1}{3}e^{3x}|_0^1 - \int_0^1 (2x - 2) \cdot \frac{1}{3}e^{3x} \, dx \quad \text{(IBP once)} \\
&= \left(\frac{2}{3}e^3 - \frac{2}{3}\right) - \int_0^1 (2x - 2) \cdot \frac{1}{3}e^{3x} \, dx \\
&= \left(\frac{2}{3}e^3 - \frac{2}{3}\right) - \left[(2x - 2) \cdot \frac{1}{9}e^{3x}|_0^1 - \int_0^1 2 \cdot \frac{1}{9}e^{3x} \, dx\right] \quad \text{(IBP again)} \\
&= \left(\frac{2}{3}e^3 - \frac{2}{3}\right) - \left[\frac{2}{9} - \left[\frac{2}{27}e^3 - \frac{2}{27}\right]\right] \\
&= \boxed{\frac{20}{27}e^3 - \frac{26}{27}}.
\end{aligned}$$

- Example: Evaluate $\int x^3 \sin(x) \, dx$.

  - Here we just need to integrate by parts repeatedly. We get

$$\begin{aligned}
\int x^3 \sin(x) \, dx &= -x^3 \cos(x) + \int 3x^2 \cos(x) \, dx \quad \text{(IBP once)} \\
&= -x^3 \cos(x) + \left[3x^2 \sin(x) - \int 6x \, \sin(x) \, dx\right] \quad \text{(IBP again)} \\
&= -x^3 \cos(x) + 3x^2 \sin(x) - \left[-6x \, \cos(x) + \int 6\cos(x) \, dx\right] \quad \text{(IBP again)} \\
&= \boxed{-x^3 \cos(x) + 3x^2 \sin(x) + 6x \, \cos(x) + 6\sin(x) + C}
\end{aligned}$$

- Example: Find $\int x^3 \sin(x^2) \, dx$.

  - First we look for a substitution. We try the argument of the sine: $u = x^2$. The differential is $du = 2x \, dx$, and we can rearrange the integral as $\int x^2 \sin(x^2) \cdot \dfrac{1}{2} \cdot (2x \, dx) = \int u \, \sin(u) \cdot \dfrac{1}{2} \, du$.
  - Now we integrate by parts, to get $-\dfrac{1}{2}u \, \cos(u) + \int \dfrac{1}{2}\cos(u) \, du = \dfrac{1}{2}\left[-u \, \cos(u) + \sin(u)\right] + C$.
  - Finally substitute back for x to get $\boxed{\dfrac{1}{2}\left[-x^2 \, \cos(x^2) + \sin(x^2)\right] + C}$.

5.4 Trigonometric Substitution

- Some kinds of integrals require a more clever sort of substitution to evaluate, one that's sort of 'backwards' from the usual way we try to do substitutions: instead of $u = f(x)$ we try $x = f(u)$ for some appropriate (trigonometric) function f. The idea is to use one of the Pythagorean relations (e.g., $\sin^2(x) + \cos^2(x) = 1$) to simplify something more complicated. As always, examples make everything clear.
- Example: Evaluate $\int \sqrt{1-x^2}\,dx$.
  - Traditional substitution – along the lines of $u = 1 - x^2$ – doesn't work like we'd hope.
  - Instead we try $x = \sin(u)$; then $dx = \cos(u)\,du$.
  - So we get $\int \sqrt{1-\sin^2(u)} \cdot \cos(u)\,du = \int \sqrt{\cos^2(u)} \cdot \cos(u)\,du = \int \cos^2(u)\,du$.
  - Remembering $\cos^2(u) = \dfrac{1+\cos(2u)}{2}$ we can evaluate the integral to get $\dfrac{u}{2} + \dfrac{\sin(2u)}{4} + C$.
  - Finally substitute back for $u = \sin^{-1}(x)$ to obtain $\boxed{\dfrac{\sin^{-1}(x)}{2} + \dfrac{\sin(2\sin^{-1}(x))}{4} + C}$. If desired, we can simplify this to the equivalent form $\boxed{\dfrac{\sin^{-1}(x)}{2} + \dfrac{x\sqrt{1-x^2}}{2} + C}$.
- Example: Evaluate $\int \dfrac{1}{(1+x^2)^2}\,dx$.
  - This time we think of the arctangent antiderivative and try $x = \tan(u)$. Then $dx = \sec^2(u)\,du$.
  - We obtain $\int \dfrac{1}{(1+\tan^2(u))^2} \cdot \sec^2(u)\,du = \int \dfrac{1}{\sec^4(u)} \cdot \sec^2(u)\,du = \int \cos^2(u)\,du$.
  - Remembering the identity $\cos^2(u) = \dfrac{1+\cos(2u)}{2}$ we can evaluate the integral to get $\dfrac{u}{2} + \dfrac{\sin(2u)}{4} + C$.
  - Finally substitute back for $u = \tan^{-1}(x)$ to obtain $\boxed{\dfrac{\tan^{-1}(x)}{2} + \dfrac{\sin(2\tan^{-1}(u))}{4} + C}$. If desired, we can simplify this to the equivalent form $\boxed{\dfrac{\tan^{-1}(x)}{2} + \dfrac{x}{2(1+x^2)} + C}$.
- Example: Evaluate $\int \dfrac{2-x}{\sqrt{4-x^2}}\,dx$.
  - We'd like to do the $x = \sin(u)$ substitution again but it doesn't quite work, since then $\sqrt{4-x^2}$ isn't nice.
  - Instead we have to try $x = 2\sin(u)$, since then $\sqrt{4-x^2} = \sqrt{4-4\sin^2(u)} = 2\cos(u)$, and we're much happier.
  - Then we have $dx = 2\cos(u)\,du$, so we get $\int \dfrac{2-2\sin(u)}{2\cos(u)} \cdot 2\cos(u)\,du = \int [2-2\sin(u)]\,du = 2u + 2\cos(u) + C$.
  - Substituting back yields $\boxed{2\sin^{-1}(x) + 2\cos(\sin^{-1}(x)) + C}$, or equivalently, $\boxed{2\sin^{-1}(x) + 2\sqrt{1-x^2} + C}$.

5.5 Partial Fractions

- Generally, it is difficult to integrate rational functions without rearranging them in some way first. (Try finding the antiderivative of $\dfrac{2x}{x^3+x^2+x+1}$ directly.)
- There is a general technique for evaluating such integrals, called partial fraction decomposition (PFD): the idea is to break down rational functions into simpler parts which we know how to integrate. To find partial fraction decompositions, follow these steps:

- Step 1: Factor the denominator.
- Step 2: Find the form of the PFD:
  - For each term $(x+a)^n$ the PFD has terms $\dfrac{C_1}{x+a} + \dfrac{C_2}{(x+a)^2} + \cdots + \dfrac{C_n}{(x+a)^n}$.
  - For each non-factorable term $(x^2+ax+b)^n$, the PFD has terms $\dfrac{C_1x+D_1}{x^2+ax+b} + \dfrac{C_2x+D_2}{(x^2+ax+b)^2} + \cdots + \dfrac{C_nx+D_n}{(x^2+ax+b)^n}$.
- Step 3: Solve for the coefficients $C_1, C_2, \ldots, C_n$.
  - The best way is to clear all denominators and then substitute 'intelligent' values for x (e.g., the roots of the linear factors).
  - Sometimes plugging in other values of x is necessary, in order to find the coefficients of the higher terms.
  - One can also employ other more clever methods, such as taking derivatives.
- Step 4: Evaluate the integral.
  - Terms of the form $\dfrac{C}{(x+a)^n}$ can be integrated directly using the Power Rule.
  - Terms of the form $\dfrac{Cx+D}{(x^2+ax+b)^n}$ should be separated further as $\dfrac{E(2x+a)}{(x^2+ax+b)^n} + \dfrac{F}{(x^2+ax+b)^n}$. The first term can then be integrated by substituting $u = x^2+ax+b$, and the second term can be integrated by completing the square as $(x+a/2)^2 + (b-a^2/4) = (x+c)^2 + d^2 = d^2\left[\left(\dfrac{x+c}{d}\right)^2 + 1\right]$, and then substituting $\tan(t) = \dfrac{x+c}{d}$.

- As ever, examples make the procedure clear.
- Example: Evaluate $\displaystyle\int \frac{1}{x^2+3x}\,dx$.
  - We factor and get $\dfrac{1}{x(x+3)}$ so we want to write $\dfrac{1}{x(x+3)} = \dfrac{A}{x} + \dfrac{B}{x+3}$.
    - Clear denominators so that $1 = A(x+3) + B(x)$.
    - Set $x = 0$ and $x = -3$ to see that $1 = 3A$ and $1 = -3B$.
  - Then $\displaystyle\int \frac{1}{x^2+3x}\,dx = \int \left[\frac{1/3}{x} - \frac{1/3}{x+3}\right] dx = \boxed{\frac{1}{3}\ln(x) - \frac{1}{3}\ln(x+3) + C}$.
- Example: Evaluate $\displaystyle\int \frac{1}{x^3+x^2}\,dx$.
  - We factor and get $\dfrac{1}{x^2(x+1)}$ so we want to write $\dfrac{1}{x^2(x+1)} = \dfrac{A}{x} + \dfrac{B}{x^2} + \dfrac{D}{x+1}$.
    - Clear denominators so that $1 = Ax(x+1) + B(x+1) + Dx^2$.
    - Set $x = 0$ to get $1 = B$.
    - Set $x = -1$ to get $1 = D$.
    - Set $x = 1$ to get $1 = 2A + 2B + D$ so that $A = -1$.
  - Then $\displaystyle\int \frac{1}{x^3+x^2}\,dx = \int \left[\frac{-1}{x} + \frac{1}{x^2} + \frac{1}{x+1}\right] dx = \boxed{-\ln(x) - \frac{1}{x} + \ln(x+1) + C}$.
- Example: Evaluate $\displaystyle\int \frac{2x}{(x+1)(x^2+1)}\,dx$.
  - We want $\dfrac{2x}{(x+1)(x^2+1)} = \dfrac{A}{x+1} + \dfrac{Bx+D}{x^2+1}$.

* Clear denominators so that $2x = A(x^2+1) + (Bx+D)(x+1)$.
* Set $x = -1$ to get $-2 = 2A$ so that $A = -1$.
* So we want $x^2 + 2x + 1 = (Bx+D)(x+1)$ so we see that we need $B = D = 1$.

- Then $\displaystyle\int \frac{2x}{(x+1)(x^2+1)}\,dx = \int \left[\frac{-1}{x+1} + \frac{x}{x^2+1} + \frac{1}{x^2+1}\right] dx = \boxed{-\ln(x+1) + \frac{1}{2}\ln(x^2+1) + \tan^{-1}(x) + C}$.

5.6 The Weierstrass Substitution

- There is one additional substitution that deserves discussion.
  - It is rarely covered at any length in modern textbooks, but it should be, since it allows one to integrate any rational function in $\sin(x)$ and $\cos(x)$.
- The substitution is $\boxed{\boxed{t = \tan\left(\frac{x}{2}\right)}}$, which is called the Weierstrass substitution, or, occasionally, "the magic substitution", since it allows one to evaluate complicated trigonometric integrals.
  - With this definition, one can check that $\cos(x) = \dfrac{1-t^2}{1+t^2}$, $\sin(t) = \dfrac{2t}{1+t^2}$, and $dx = \dfrac{2}{1+t^2}\,dt$.
  - Then as we can see, the trigonometric integral $\displaystyle\int \frac{p(\sin(x),\cos(x))}{q(\sin(x),\cos(x))}\,dx$ becomes the integral $\displaystyle\int \frac{p\left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2}\right)}{q\left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2}\right)} \frac{2}{1+t^2}$ which is a rational function of t (though complicated).
  - The magic substitution can also be used to reduce any problem involving trigonometric identities in rational multiples of a single variable x to a finite computation with rational functions. (In practice, this is not as useful as it might seem, because the polynomial algebra is usually harder than using other techniques like the angle-addition formulas.)
- Example: Find $\displaystyle\int \frac{1}{2+\cos(x)}\,dx$.
  - We set $t = \tan\left(\frac{x}{2}\right)$, with $\cos(x) = \dfrac{1-t^2}{1+t^2}$ and $dx = \dfrac{2}{1+t^2}\,dt$, to obtain

$$\begin{aligned}\int \frac{1}{2+\cos(x)}\,dx &= \int \frac{1}{2+\frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2}\,dt \\ &= \int \frac{2}{2(1+t^2)+(1-t^2)}\,dt \\ &= \int \frac{2}{3+t^2}\,dt = \int \frac{2}{3} \cdot \frac{1}{1+(t/\sqrt{3})^2}\,dt\end{aligned}$$

  - In this new integral we set $u = t/\sqrt{3}$ with $du = dt/\sqrt{3}$ to obtain $\displaystyle\int \frac{2}{3} \cdot \frac{\sqrt{3}}{1+u^2}\,du = \frac{2}{\sqrt{3}}\tan^{-1}(u) + C$.
  - Substituting back for t and then x yields the answer as $\boxed{\dfrac{2}{\sqrt{3}}\tan^{-1}\left(\dfrac{1}{\sqrt{3}}\tan\left(\dfrac{x}{2}\right)\right) + C}$.

5.7 Improper Integration

- Sometimes we like to integrate to infinity, by which we mean, take the limit as a limit of integration becomes arbitrarily large.

- In other words, we write $\int_a^\infty f(x)\,dx$ as shorthand for $\lim_{q\to\infty}\int_a^q f(x)\,dx$.

- Other times, we like to integrate through a 'singularity' of a function – that is, through a point where a function is undefined because it blows up: for example, $x=0$ for the function $f(x)=1/x$.
  - Again, we will write $\int_0^1 \frac{1}{x}\,dx$ as shorthand for $\lim_{q\to 0^+}\int_q^1 \frac{1}{x}\,dx$.

- Integrals with either of these two kinds of 'bad behaviors' are called improper integrals. (Perhaps the terminology stems from the fact that trying to evaluate such integrals is not something done in polite company.)

- Typically we will be interested in asking (i) if the integral actually converges to a finite value, and (ii) if it does, what the value is.

- There are roughly two ways to solve problems like this:
  - Method 1: Find the indefinite integral, and then evaluate the limit. If the limit is hard, try rewriting the function to simplify the limit.
  - Method 2: Compare the integrand to another function whose integral is easier to evaluate. We are aided by the following theorem:
    - Theorem (Comparison Test for Integrals): If $0<f(x)<g(x)$ for all x in the interval $[a,b]$ where one or both of a and b can be infinite, then if $\int_a^b f(x)\,dx$ diverges, so does $\int_a^b g(x)\,dx$. If $\int_a^b g(x)\,dx$ converges, then so does $\int_a^b f(x)\,dx$.
    - The idea behind the theorem is to say that if $0<f(x)<g(x)$ then $\int_a^b f(x)\,dx < \int_a^b g(x)\,dx$, and so if the $f(x)$-integral goes to ∞ then so must the $g(x)$-integral, and conversely if the $g(x)$-integral stays finite then so must the $f(x)$-integral.

- Example: Evaluate $\int_0^\infty e^{-x}\,dx$.
  - We just evaluate to see $\int_0^\infty e^{-x}\,dx = \lim_{q\to\infty}(-e^{-x})|_{x=0}^q = \lim_{q\to\infty}\left[-e^{-q}+1\right] = \boxed{1}$.

- Example: Evaluate $\int_1^\infty \frac{1}{x^2+x}\,dx$.
  - We do partial fractions to see $\int \frac{1}{x^2+x}\,dx = \ln(x)-\ln(x+1)+C$.
  - Now to take the limit as $x\to\infty$ is not so easy, unless we notice that we can rewrite the indefinite integral as $\ln\left(\frac{x}{x+1}\right)$.
  - Then we have $\int_1^\infty \frac{1}{x^2+x}\,dx = \lim_{q\to\infty}\left[\ln\left(\frac{x}{x+1}\right)\right]|_{x=1}^q = \lim_{q\to\infty}\left[\ln\left(\frac{x}{x+1}\right)-\ln(\frac{1}{2})\right] = \boxed{\ln(2)}$.
  - If the problem had only asked about convergence, we could have observed that $x^2+x>x^2$ for positive x, so $\frac{1}{x^2+x}<\frac{1}{x^2}$. Then the Comparison Test would have said that, because $\int_1^\infty \frac{1}{x^2}\,dx = -\frac{1}{x}|_{x=1}^\infty = 1$ converges, then so does $\int_1^\infty \frac{1}{x^2+x}\,dx$. However, the Comparison Test doesn't help us in computing the actual value.

- Example: Determine whether the integral $\int_0^\infty \frac{1}{(x+1)(x+2)(x+3)(x+4)}\,dx$ converges.
  - Doing partial fractions is too much work (although it does give the correct answer, if the trick in the previous example is used repeatedly).
  - But it's easier to notice that $\frac{1}{x+4}<\frac{1}{x+3}<\frac{1}{x+2}<\frac{1}{x+1}$.

- Then $\int_0^q \frac{1}{(x+1)(x+2)(x+3)(x+4)}\,dx < \int_0^q \frac{1}{(x+1)^4}\,dx = -\frac{1}{4}(x+1)^{-3}|_{x=0}^{q}$. As $q \to \infty$ this remains finite (in fact, it is $\frac{1}{4}$).
- Therefore this integral $\boxed{\text{converges}}$.

- Example: Determine whether the integral $\int_0^1 \frac{1}{x}\,dx$ converges.
  - We integrate to get $\int_q^1 \frac{1}{x}\,dx = -\ln(q)$.
  - So as $q \to 0^+$ we see that this diverges to $-\infty$.
  - Therefore this integral $\boxed{\text{diverges to } -\infty}$.

Well, you're at the end of my handout. Hope it was helpful.